EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Edge-Case Analysis — long mathematics and missing values

Physics + Chemistry + Biology · NEET · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	TOTAL MARKS
Aarav Sharma	21NEET114	2026-07-05	60

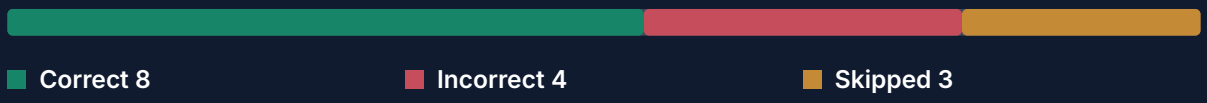
TOTAL MARKS OBTAINED

28

/ 60 marks

8 correct | 4 incorrect | 3 skipped





Correct 8 · Incorrect 4 · Skipped 3

SCORE	ACCURACY	RESPONSE DISTRIBUTION
60%	—%	8 correct | 4 incorrect | 3 skipped

PUBLISHED PERFORMANCE BRIEF

Synthetic QA fixture: long mathematical explanations, malformed TeX, missing optional values and 32 question rows. Not a student report.

FAR / QA-EDGE-01 · QA-EDGE-01 · Source fingerprint 3DF6B6E8 · 2026-11-02T20:39:58.849Z

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

32 questions in the paper



- Correct 8
- Incorrect 4
- Skipped 3

Mistake distribution

Count by published classification

Calculation Error	2
Unit Error	1
Conceptual Error	1

Seven source classifications; rows are measured independently.

Difficulty analysis

Questions and published accuracy

Easy	— Q · 80%
Medium	— Q · 50%
Hard	— Q · 0%

Accuracy values are reproduced from the payload.

Subject-wise performance

Accuracy by subject

Physics	68%
Chemistry	52%

Physics 72 / 100 Chemistry 41 / 100

Important performance signals

GRADE	RANK	PERCENTILE	BATCH RANK	COHORT SIZE	OVERALL SPEED	CONFIDENCE INDEX	READINESS SCORE
B	—	—	—	—	—	66.7	60

HEALTH SCORE	NEGATIVE MARKS	GUESS ATTEMPTS	ATTEMPT RATIO	TOTAL QUESTIONS	SUCCESS RATE	BADGE
60	2	0	80%	32	66.7%	Completed

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	72 / 100	68%	—	Medium	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
Chemistry	41 / 100	52%	—	Hard	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	—	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	—
Chemistry	Chemical Kinetics	—	—

Chapter mastery · grouped by subject

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	MASTERY
Physics					
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	48%	24	18m 30s	Weak
Chemistry					
Chemical Kinetics	Chemistry	76%	18	12m	Strong

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SUBJECT	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics				
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	61%	18m 30s	61%
Chemistry				
Chemical Kinetics	Chemistry	39%	12m	39%

Time-management metrics

AVERAGE / QUESTION	NAVIGATION COUNT	MANAGEMENT SCORE	FASTEST QUESTION
1.68	0	60	Q1

FASTEST TIME	SLOWEST QUESTION	SLOWEST TIME
0.87	Q15	2.5

Sectional allocation and efficiency

SECTION	ALLOC.	USED	EFF.
Physics	30	5.5	18.3
Chemistry	30	8.4	28
Biology	30	11.3	37.7

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Benchmark scores

Previous test —
Previous month —
Batch average —
Institute average —
Top performer —

Current score 28 / 60

Comparison register

BENCHMARK	SCORE	PCT.	GRADE	REFERENCE
Previous test	—	—	—	—
Previous month	—	—	—	—
Batch average	—	—	—	—
Institute average	—	—	—	—
Top performer	—	—	—	—

Projection and readiness

Score trajectory

Expected improvement — · Risk —

JEE PERCENTILE	NEET SCORE	EXPECTED AIR
—	36	—

TARGET PROBABILITY	RISK LEVEL	EXPECTED IMPROVEMENT
—	—	—

Trajectory values

EXAMINATION / PROJECTION	SCORE

READINESS SCORE	CONFIDENCE INDEX	HEALTH SCORE	OVERALL ACCURACY	SUCCESS RATE	BADGE
60	66.7	60	—%	66.7%	Completed

05 QUESTION INTELLIGENCE

Question-by-question analysis

32 question-review rows supplied. Missing optional values are shown as —; full source explanations are never shortened.

REVIEW LEGEND

Correct · Incorrect · Skipped Each question presents its identity, classifications, answer comparison, marks, time, mistake category and complete solution.

Q1

Q1 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	Numerical	Correct

QUESTION / OPTIONS

Question: Derive case 1: $\frac{d}{dx}\left(x^2\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{3} k^2$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	—	—

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q2

Q2 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion

DIFFICULTY	TYPE	STATUS
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 2: $\frac{d}{dx}\left(x^3\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{4} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	1m 10s

Mistake category · Conceptual gap

EXPLANATION

Using

Q2 · solution continued

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q3 Q3 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 3: $\frac{d}{dx}\left(x^4\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{5} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	1m 32s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\ a + d &= b + c + e + f \end{aligned}$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q4

Q4 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 4: $\frac{d}{dx}\left(x^5\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{6} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	1m 55s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q5

Q5 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Derive case 5: $\frac{d}{dx}\left(x^6\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{7} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	-1	2m 17s

Mistake category · Conceptual gap

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q6

Q6 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 6: $\frac{d}{dx}\left(x^7\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{8} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	2m 39s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q7 Q7 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 7: $\frac{d}{dx}\left(x^8\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{9} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	3m 01s

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q8 Q8 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 8: $\frac{d}{dx}\left(x^9\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{10} k^2$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	—	—

Mistake category · Conceptual gap

EXPLANATION

FLAGGED SOURCE — do not correct:

Q8 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q9 Q9 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Hard	Numerical	Skipped

QUESTION / OPTIONS

Question: Derive case 9: $\frac{d}{dx}\left(x^{10}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{11} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	3m 46s

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q10

Q10 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 10: $\frac{d}{dx}\left(x^{11}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{12} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	48s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q11

Q11 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 11: $\frac{d}{dx}\left(x^{12}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{13} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	-1	1m 10s

Mistake category · Conceptual gap

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q12

Q12 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 12: $\frac{d}{dx}\left(x^{13}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{14} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	1m 32s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q13

Q13 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Easy	Numerical	Correct

QUESTION / OPTIONS

Question: Derive case 13: $\frac{d}{dx}\left(x^{14}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{15} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	1m 55s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q14

Q14 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 14: $\frac{d}{dx}\left(x^{15}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{16} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	2m 17s

Mistake category · Conceptual gap

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q15 Q15 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
—	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 15: $\frac{d}{dx}\left(x^{16}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{17} k^2$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	—	—

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q16 Q16 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 16: $\frac{d}{dx}\left(x^{17}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{18} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	3m 01s

EXPLANATION

FLAGGED SOURCE — do not correct:

Q16 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q17

Q17 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Derive case 17: $\frac{d}{dx}\left(x^{18}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{19} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	-1	3m 23s

Mistake category · Conceptual gap

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q17 · solution continued

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q18 Q18 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 18: $\frac{d}{dx}\left(x^{19}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{20} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	3m 46s

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q19

Q19 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 19: $\frac{d}{dx}\left(x^{20}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{21} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	48s

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q20

Q20 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	TYPE	STATUS
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 20: $\frac{d}{dx}\left(x^{21}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{22} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	-1	1m 10s

Mistake category · Conceptual gap

EXPLANATION

FLAGGED SOURCE — do not correct:

Q20 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q21 Q21 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Hard	Numerical	Skipped

QUESTION / OPTIONS

Question: Derive case 21: $\frac{d}{dx}\left(x^{22}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{23} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	1m 32s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q21 · solution continued

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q22 Q22 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	TYPE	STATUS
—	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 22: $\frac{d}{dx}\left(x^{23}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{24} k^2$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	—	—

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q23 Q23 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	TYPE	STATUS
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 23: $\frac{d}{dx}\left(x^{24}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{25} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	-1	2m 17s

Mistake category · Conceptual gap

EXPLANATION

A multi-line derivation:

Q23 · solution continued

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q24 Q24 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 24: $\frac{d}{dx}\left(x^{25}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{26} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	2m 39s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q25

Q25 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Easy	Numerical	Correct

QUESTION / OPTIONS

Question: Derive case 25: $\frac{d}{dx}\left(x^{26}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{27} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	3m 01s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q26

Q26 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 26: $\frac{d}{dx}\left(x^{27}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{28} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	3m 23s

Mistake category · Conceptual gap

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q27 Q27 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 27: $\frac{d}{dx}\left(x^{28}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{29} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	3m 46s

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q28 Q28 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 28: $\frac{d}{dx}\left(x^{29}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{30} k^2$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	48s

EXPLANATION

FLAGGED SOURCE — do not correct:

Q28 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q29

Q29 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
—	Numerical	Incorrect

QUESTION / OPTIONS

Question: Derive case 29: $\frac{d}{dx}\left(x^{30}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{31} k^2$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	—	—

Mistake category · Conceptual gap

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

MISTAKE NOTE

Compare the boundary conditions before substituting.

Q30

Q30 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	MCQ	Skipped

QUESTION / OPTIONS

Question: Derive case 30: $\frac{d}{dx}\left(x^{31}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{32} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	1m 32s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q31

Q31 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Derive case 31: $\frac{d}{dx}\left(x^{32}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{33} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	1m 55s

EXPLANATION

A multi-line derivation:

$$a = b + c + \frac{1}{2}$$
$$d = e + f - \sqrt{\lambda}$$
$$\therefore\ a + d = b + c + e + f$$

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

Q32

Q32 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	MCQ	Incorrect

QUESTION / OPTIONS

Question: Derive case 32: $\frac{d}{dx}\left(x^{33}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{34} k^2$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	-1	2m 17s

Mistake category · Conceptual gap

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

SOLUTION / HINT

Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.

MISTAKE NOTE

Compare the boundary conditions before substituting.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	High	This week

Topic-level accuracy

TOPIC	SUBJECT	ACC.	LEVEL
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Physics	41%	Weak
Rate laws	Chemistry	82%	Strong

Mistake evidence

Q	SOURCE DETAIL
Q3	Calculation Error · Plane mirror
Q6	Calculation Error · Iodoform test
Q9	Unit Error · Reducing sugars
Q12	Conceptual Error · Monohybrid cross

Weak chapters

NO.	WEAK CHAPTER
1	**Optics**
2	**Modern Physics**
3	**Organic Chemistry**
4	**Physical Chemistry**
5	**Genetics & Evolution**
6	**Ecology**

Subject-wise revision sections

Priority topics grouped by subject

SUBJECT	PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics				
Physics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	41%	Critical
Also revise				
Also revise	Plane mirror	—	—	Revise
Also revise	Photon energy	—	—	Revise
Also revise	Iodoform test	—	—	Revise
Also revise	Reducing sugars	—	—	Revise
Also revise	Spontaneity criteria	—	—	Revise
Also revise	Monohybrid cross	—	—	Revise
Also revise	Energy pyramid	—	—	Revise

Recommended next actions

Weak topics / published list

NO.	WEAK TOPIC
1	Plane mirror
2	Photon energy
3	Iodoform test
4	Reducing sugars
5	Spontaneity criteria
6	Monohybrid cross
7	Energy pyramid

PUBLISHED PERFORMANCE BRIEF

Synthetic QA fixture: long mathematical explanations, malformed TeX, missing optional values and 32 question rows. Not a student report.

08 DATA INTEGRITY

Source data register

Source data register · 338 canonical payload values

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
attempt.id MOCK-NEET-2026-07	attempt.examId EA-NEET-FULL-01	attempt.examTitle NEET UG · Full Mock 01	attempt.shortTitle NEET UG Mock 01
attempt.examMode Competitive	attempt.examFamily NEET	attempt.submittedAt 2026-07-05T10:30:00	attempt.startedAt 2026-07-05T07:30:00
attempt.studentName Aarav Sharma	attempt.roll 21NEET114	attempt.scoring.score 36	attempt.scoring.maxScore 60
attempt.scoring.correct 10	attempt.scoring.incorrect 2	attempt.scoring.skipped 3	attempt.summary.accuracy 83.3
attempt.exam.subject Physics + Chemistry + Biology	attempt.exam.durationMinutes 30	attempt.exam.marksPerQuestion 4	attempt.exam.negativeMarksPerQuestion 1
attempt.exam.difficulty Mixed	attempt.exam.format 15 MCQs · 30 minutes · +4 marks · −1 per incorrect answer	attempt.exam.instructions[0] This is a practice mock in NEET UG format with negative marking.	attempt.exam.instructions[1] Each question carries 4 marks; every incorrect answer deducts 1 mark.
attempt.exam.instructions[2] Unanswered questions receive 0 marks — do not guess blindly.	attempt.exam.instructions[3] Use the question navigator to jump between questions; answers are auto-saved.	attempt.exam.demoProfile.strongChapters[0] Human Physiology	attempt.exam.demoProfile.weakChapters[0] Modern Physics
attempt.exam.demoProfile.weakChapters[1] Physical Chemistry	attempt.exam.demoProfile.baseAccuracy 0.72	attempt.exam.questions[0].id Q01	attempt.exam.questions[0].subject Physics
attempt.exam.questions[0].chapter Mechanics	attempt.exam.questions[0].topic Unit conversion	attempt.exam.questions[0].difficulty Easy	attempt.exam.questions[0].question A car moving at 54 km/h has a speed of...
attempt.exam.questions[0].options[0] 15 m/s	attempt.exam.questions[0].options[1] 20 m/s	attempt.exam.questions[0].options[2] 25 m/s	attempt.exam.questions[0].options[3] 10 m/s
attempt.exam.questions[0].correctAnswer 0	attempt.exam.questions[0].type MCQ	attempt.exam.questions[1].id Q02	attempt.exam.questions[1].chapter Current Electricity

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
attempt.exam.questions[1].topic Parallel resistors	attempt.exam.questions[1].difficulty Medium	attempt.exam.questions[1].question Three 6 Ω resistors connected in parallel have an equivalent resistance of...	attempt.exam.questions[1].options[0] 18 Ω
attempt.exam.questions[1].options[1] 6 Ω	attempt.exam.questions[1].options[2] 2 Ω	attempt.exam.questions[1].options[3] 3 Ω	attempt.exam.questions[2].id Q03
attempt.exam.questions[2].chapter Optics	attempt.exam.questions[2].topic Plane mirror	attempt.exam.questions[2].question The image formed by a plane mirror is always...	attempt.exam.questions[2].options[0] Virtual and erect
attempt.exam.questions[2].options[1] Real and inverted	attempt.exam.questions[2].options[2] Virtual and inverted	attempt.exam.questions[2].options[3] Real and erect	attempt.exam.questions[3].id Q04
attempt.exam.questions[3].chapter Thermodynamics	attempt.exam.questions[3].topic Isothermal process	attempt.exam.questions[3].question In an isothermal process for an ideal gas, the change in internal energy is...	attempt.exam.questions[3].options[0] Zero
attempt.exam.questions[3].options[1] Maximum	attempt.exam.questions[3].options[2] Negative	attempt.exam.questions[3].options[3] Equal to the work done	attempt.exam.questions[4].id Q05
attempt.exam.questions[4].topic Photon energy	attempt.exam.questions[4].difficulty Hard	attempt.exam.questions[4].question The energy of a photon of frequency 5×10^{14} Hz (h = 6.6×10^{-34} J·s) is about...	attempt.exam.questions[4].options[0] 3.3×10^{-19} J
attempt.exam.questions[4].options[1] 6.6×10^{-19} J	attempt.exam.questions[4].options[2] 1.1×10^{-19} J	attempt.exam.questions[4].options[3] 9.9×10^{-19} J	attempt.exam.questions[5].id Q06
attempt.exam.questions[5].subject Chemistry	attempt.exam.questions[5].chapter Organic Chemistry	attempt.exam.questions[5].topic Iodoform test	attempt.exam.questions[5].question Which of the following gives a positive iodoform test?
attempt.exam.questions[5].options[0] Acetone	attempt.exam.questions[5].options[1] Methanol	attempt.exam.questions[5].options[2] Formaldehyde	attempt.exam.questions[5].options[3] Acetic acid
attempt.exam.questions[6].id Q07	attempt.exam.questions[6].chapter Inorganic Chemistry	attempt.exam.questions[6].topic Chlorophyll metal	attempt.exam.questions[6].question The metal present at the centre of the chlorophyll molecule is...
attempt.exam.questions[6].options[0] Magnesium	attempt.exam.questions[6].options[1] Iron	attempt.exam.questions[6].options[2] Calcium	attempt.exam.questions[6].options[3] Zinc
attempt.exam.questions[7].id Q08	attempt.exam.questions[7].topic pH scale	attempt.exam.questions[7].question The pH of a 10^{-3} M HCl solution is...	attempt.exam.questions[7].options[1] 11
attempt.exam.questions[7].options[2] 7	attempt.exam.questions[8].id Q09	attempt.exam.questions[8].topic Reducing sugars	attempt.exam.questions[8].question Which of the following is a reducing sugar?

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
attempt.exam.questions[8].options[0] Sucrose	attempt.exam.questions[8].options[1] Glucose	attempt.exam.questions[8].options[2] Starch	attempt.exam.questions[8].options[3] Cellulose
attempt.exam.questions[9].id Q10	attempt.exam.questions[9].topic Spontaneity criteria	attempt.exam.questions[9].question For a reaction with ΔH positive and ΔS positive, spontaneity is favoured at...	attempt.exam.questions[9].options[0] High temperature
attempt.exam.questions[9].options[1] Low temperature	attempt.exam.questions[9].options[2] All temperatures	attempt.exam.questions[9].options[3] No temperature	attempt.exam.questions[10].id Q11
attempt.exam.questions[10].subject Biology	attempt.exam.questions[10].topic Blood cells	attempt.exam.questions[10].question Which blood cells are primarily responsible for immune defence?	attempt.exam.questions[10].options[0] Red blood cells
attempt.exam.questions[10].options[1] White blood cells	attempt.exam.questions[10].options[2] Platelets	attempt.exam.questions[10].options[3] Plasma cells only	attempt.exam.questions[11].id Q12
attempt.exam.questions[11].chapter Genetics & Evolution	attempt.exam.questions[11].topic Monohybrid cross	attempt.exam.questions[11].question In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is...	attempt.exam.questions[11].options[0] 3 : 1
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attempt.exam.questions[12].chapter Cell Biology	attempt.exam.questions[12].topic Mitochondria	attempt.exam.questions[12].question Which organelle is known as the powerhouse of the cell?	attempt.exam.questions[12].options[0] Ribosome
attempt.exam.questions[12].options[1] Mitochondrion	attempt.exam.questions[12].options[2] Lysosome	attempt.exam.questions[12].options[3] Golgi apparatus	attempt.exam.questions[13].id Q14
attempt.exam.questions[13].chapter Plant Physiology	attempt.exam.questions[13].topic Transpiration	attempt.exam.questions[13].question The loss of water as vapour from plants, mainly through stomata, is called...	attempt.exam.questions[13].options[1] Guttation
attempt.exam.questions[13].options[2] Respiration	attempt.exam.questions[13].options[3] Osmosis	attempt.exam.questions[14].id Q15	attempt.exam.questions[14].chapter Ecology
attempt.exam.questions[14].topic Energy pyramid	attempt.exam.questions[14].question The pyramid of energy in any ecosystem is always...	attempt.exam.questions[14].options[0] Upright	attempt.exam.questions[14].options[1] Inverted
attempt.exam.questions[14].options[2] Spindle shaped	attempt.exam.questions[14].options[3] Linear	attempt.exam.academicYear 2026-27	attempt.exam.venue Main hall
attempt.exam.seatNumber A-114	attempt.questionAttempts[0].questionId EA-NEET-FULL-01-Q01	attempt.questionAttempts[0].evaluation.isCorrect true	attempt.questionAttempts[0].evaluation.isSkipped false
attempt.questionAttempts[0].timing.elapsedSeconds 52	attempt.questionAttempts[1].questionId EA-NEET-FULL-01-Q02	attempt.questionAttempts[1].timing.elapsedSeconds 59	attempt.questionAttempts[2].questionId EA-NEET-FULL-01-Q03

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
attempt.questionAttempts[2].evaluation.marksAwarded -1	attempt.questionAttempts[2].timing.elapsedSeconds 66	attempt.questionAttempts[3].questionId EA-NEET-FULL-01-Q04	attempt.questionAttempts[3].timing.elapsedSeconds 73
attempt.questionAttempts[4].questionId EA-NEET-FULL-01-Q05	attempt.questionAttempts[4].questionNumber 5	attempt.questionAttempts[4].response.status skipped	attempt.questionAttempts[4].timing.elapsedSeconds 80
attempt.questionAttempts[5].questionId EA-NEET-FULL-01-Q06	attempt.questionAttempts[5].questionNumber 6	attempt.questionAttempts[5].timing.elapsedSeconds 87	attempt.questionAttempts[6].questionId EA-NEET-FULL-01-Q07
attempt.questionAttempts[6].timing.elapsedSeconds 94	attempt.questionAttempts[7].questionId EA-NEET-FULL-01-Q08	attempt.questionAttempts[7].questionNumber 8	attempt.questionAttempts[7].timing.elapsedSeconds 101
attempt.questionAttempts[8].questionId EA-NEET-FULL-01-Q09	attempt.questionAttempts[8].questionNumber 9	attempt.questionAttempts[8].timing.elapsedSeconds 108	attempt.questionAttempts[9].questionId EA-NEET-FULL-01-Q10
attempt.questionAttempts[9].timing.elapsedSeconds 115	attempt.questionAttempts[10].questionId EA-NEET-FULL-01-Q11	attempt.questionAttempts[10].timing.elapsedSeconds 122	attempt.questionAttempts[11].questionId EA-NEET-FULL-01-Q12
attempt.questionAttempts[11].questionNumber 12	attempt.questionAttempts[11].timing.elapsedSeconds 129	attempt.questionAttempts[12].questionId EA-NEET-FULL-01-Q13	attempt.questionAttempts[12].questionNumber 13
attempt.questionAttempts[12].timing.elapsedSeconds 136	attempt.questionAttempts[13].questionId EA-NEET-FULL-01-Q14	attempt.questionAttempts[13].questionNumber 14	attempt.questionAttempts[13].timing.elapsedSeconds 143
attempt.questionAttempts[14].questionId EA-NEET-FULL-01-Q15	attempt.questionAttempts[14].questionNumber 15	attempt.questionAttempts[14].timing.elapsedSeconds 150	examId QA-EDGE-01
examTitle QA Edge-Case Analysis — long mathematics and missing values	chapters[0].accuracy 48	chapters[0].marks 24	chapters[0].time 18.5
chapters[0].mastery Weak	chapters[1].chapter Chemical Kinetics	chapters[1].accuracy 76	chapters[1].marks 18
chapters[1].attempted 90	chapters[1].mastery Strong	source exam-agent	hero.score 28
hero.grade B	hero.confidenceIndex 66.7	hero.badge Completed	meta.date 2026-07-05
meta.resultStatus Declared	cognitiveQuadrants.medianTimeMinutes 1.68	timeIntelligence.distribution[0].used 5.5	timeIntelligence.distribution[0].efficiency 18.3
timeIntelligence.distribution[1].used 8.4	timeIntelligence.distribution[2].used 11.3	timeIntelligence.distribution[2].efficiency 37.7	timeIntelligence.fastestQuestion.q Q1
timeIntelligence.fastestQuestion.time 0.87	timeIntelligence.slowestQuestion.time 2.5	subjects[0].score 72	subjects[0].maxMarks 100
subjects[0].accuracy 68	subjects[0].time 42.5	subjects[0].strongAreas[0] Rotational Motion	subjects[1].score 41
subjects[1].time 38	topics[1].topic Rate laws	topics[1].mastery 82	questionReview[0].type Numerical
questionReview[0].status Correct	questionReview[1].q Q2	questionReview[1].topic Projectile motion	questionReview[1].status Incorrect
questionReview[1].time 1.17	questionReview[1].options[0] 2 s	questionReview[1].options[1] $\sqrt{3}$ s	questionReview[1].options[2] 4 s

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
questionReview[1].options[3] A very long alternative answer that must wrap without clipping the card	questionReview[1].mistakeCategory Conceptual gap	questionReview[1].mistakeHint Compare the boundary conditions before substituting.	questionReview[2].q Q3
questionReview[2].time 1.54	questionReview[3].q Q4	questionReview[3].time 1.91	questionReview[4].q Q5
questionReview[4].time 2.28	questionReview[5].q Q6	questionReview[5].time 2.65	questionReview[6].q Q7
questionReview[6].time 3.02	questionReview[7].q Q8	questionReview[8].q Q9	questionReview[8].time 3.76
questionReview[9].time 0.8	questionReview[15].q Q16	questionReview[16].q Q17	questionReview[16].time 3.39
questionReview[17].q Q18	questionReview[18].q Q19	questionReview[19].q Q20	questionReview[20].q Q21
questionReview[21].q Q22	questionReview[22].q Q23	questionReview[23].q Q24	questionReview[24].q Q25
questionReview[25].q Q26	questionReview[26].q Q27	questionReview[27].q Q28	questionReview[28].q Q29
questionReview[29].q Q30	questionReview[30].q Q31	questionReview[31].q Q32	mistakeList[0].mistakeCategory Calculation Error
mistakeList[0].detail Calculation Error · Plane mirror	mistakeList[1].detail Calculation Error · Iodoform test	mistakeList[2].mistakeCategory Unit Error	mistakeList[2].detail Unit Error · Reducing sugars
mistakeList[3].mistakeCategory Conceptual Error	mistakeList[3].detail Conceptual Error · Monohybrid cross	difficulty[1].accuracy 50	difficulty[1].time 13.6
difficulty[2].time 3.2	comparison.previousTest.label Previous test	comparison.previousMonth.label Previous month	comparison.batchAverage.label Batch average
comparison.instituteAverage.label Institute average	comparison.topPerformer.label Top performer	recommendations.priorityRevision[0].timeframe This week	recommendations.priorityRevision[0].priority High

Extended source values · 46 verbatim entries

SOURCE / VALUE	SOURCE / VALUE
attempt.exam.description Full-syllabus NEET UG style mock — 5 Physics, 5 Chemistry and 5 Biology questions with +4 / −1 marking.	attempt.exam.instructions[4] The AI Exam Agent analyses only your interactions — timing, answers and revisits. No camera, microphone or device monitoring is used.
chapters[0].chapter Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	hero.aiSummary Synthetic QA fixture: long mathematical explanations, malformed TeX, missing optional values and 32 question rows. Not a student report.
subjects[0].weakAreas[0] Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	questionReview[0].text Derive case 1: $\frac{d}{dx}\left(x^{2}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{3} k^2$.
questionReview[0].explanation The integral reduces directly: $$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$ The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.	questionReview[1].text Derive case 2: $\frac{d}{dx}\left(x^{3}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{4} k^2$.

SOURCE / VALUE	SOURCE / VALUE
questionReview[1].hint Review Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions before attempting the next variation.	questionReview[1].explanation Using $$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$ and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\ce{H2SO4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.
questionReview[2].text Derive case 3: $\frac{d}{dx}\left(x^{4}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{5} k^2$.	questionReview[2].explanation A multi-line derivation: $$\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\\end{aligned}$$ with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty} (1+1/n)^n = e$.
questionReview[3].text Derive case 4: $\frac{d}{dx}\left(x^{5}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{6} k^2$.	questionReview[3].explanation FLAGGED SOURCE — do not correct: $\frac{1}{2$ is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced: $$\int_0^1 \frac{x}{\sqrt{1-x^2}$$. Neither fragment is guessed or silently repaired.
questionReview[4].text Derive case 5: $\frac{d}{dx}\left(x^{6}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{7} k^2$.	questionReview[5].text Derive case 6: $\frac{d}{dx}\left(x^{7}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{8} k^2$.
questionReview[6].text Derive case 7: $\frac{d}{dx}\left(x^{8}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{9} k^2$.	questionReview[7].text Derive case 8: $\frac{d}{dx}\left(x^{9}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{10} k^2$.
questionReview[8].text Derive case 9: $\frac{d}{dx}\left(x^{10}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{11} k^2$.	questionReview[9].text Derive case 10: $\frac{d}{dx}\left(x^{11}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{12} k^2$.
questionReview[10].text Derive case 11: $\frac{d}{dx}\left(x^{12}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{13} k^2$.	questionReview[11].text Derive case 12: $\frac{d}{dx}\left(x^{13}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{14} k^2$.
questionReview[12].text Derive case 13: $\frac{d}{dx}\left(x^{14}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{15} k^2$.	questionReview[13].text Derive case 14: $\frac{d}{dx}\left(x^{15}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{16} k^2$.
questionReview[14].text Derive case 15: $\frac{d}{dx}\left(x^{16}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{17} k^2$.	questionReview[15].text Derive case 16: $\frac{d}{dx}\left(x^{17}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{18} k^2$.
questionReview[16].text Derive case 17: $\frac{d}{dx}\left(x^{18}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{19} k^2$.	questionReview[17].text Derive case 18: $\frac{d}{dx}\left(x^{19}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{20} k^2$.
questionReview[18].text Derive case 19: $\frac{d}{dx}\left(x^{20}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{21} k^2$.	questionReview[19].text Derive case 20: $\frac{d}{dx}\left(x^{21}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{22} k^2$.
questionReview[20].text Derive case 21: $\frac{d}{dx}\left(x^{22}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{23} k^2$.	questionReview[21].text Derive case 22: $\frac{d}{dx}\left(x^{23}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{24} k^2$.
questionReview[22].text Derive case 23: $\frac{d}{dx}\left(x^{24}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{25} k^2$.	questionReview[23].text Derive case 24: $\frac{d}{dx}\left(x^{25}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{26} k^2$.
questionReview[24].text Derive case 25: $\frac{d}{dx}\left(x^{26}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{27} k^2$.	questionReview[25].text Derive case 26: $\frac{d}{dx}\left(x^{27}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{28} k^2$.
questionReview[26].text Derive case 27: $\frac{d}{dx}\left(x^{28}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{29} k^2$.	questionReview[27].text Derive case 28: $\frac{d}{dx}\left(x^{29}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{30} k^2$.

SOURCE / VALUE	SOURCE / VALUE
questionReview[28].text Derive case 29: $\frac{d}{dx}\left(x^{30}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{31} k^2$.	questionReview[29].text Derive case 30: $\frac{d}{dx}\left(x^{31}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{32} k^2$.
questionReview[30].text Derive case 31: $\frac{d}{dx}\left(x^{32}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{33} k^2$.	questionReview[31].text Derive case 32: $\frac{d}{dx}\left(x^{33}\sqrt{1-x^2}\right)$ and evaluate $\sum_{k=1}^{34} k^2$.
mistakeList[0].hint Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Plane mirror.	mistakeList[1].hint Your approach looks right but a computational slip derailed the answer. Double-check each arithmetic step and watch for sign/factor errors. Topic: Iodoform test.
mistakeList[2].hint The numerical approach is likely correct but the unit or scale used in the final step is off. Verify unit conversions and dimensional consistency.	mistakeList[3].hint Review the underlying concept — the core idea behind this topic seems unclear. Re-read the definition and work through examples before retrying.

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End of report · FAR / QA-EDGE-01

Missing values are shown as —.